United States Patent Application Publication	20250271735
Kind Code	A1
Publication Date	August 28, 2025
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IMAGE CAPTURING APPARATUS THAT IS IMPROVED IN OPERABILITY WHILE REDUCING SIZE OF GRIP

Abstract

An image capturing apparatus including an image capturing apparatus body and a grip rotatably attached to a side surface of the body. The grip includes a grasping portion for being grasped by a user, a first, a second, and a third operation keys for being operated by the user, a lower grip exterior portion covering the first operation key at least from the body, and an upper grip exterior portion covering the third operation key at least from the body. When the grip is in a position where the first and second operation keys are upward of the grasping portion, the third operation key is upward of a line segment connecting between the first and second operation keys, and at the same time, the upper grip exterior portion is more remote from the body than the lower grip exterior portion.

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Family ID: **1000008493138**

Appl. No.: **19/056141**

Filed: **February 18, 2025**

Foreign Application Priority Data

JP	2024-026566	Feb. 26, 2024

Publication Classification

Int. Cl.: **G03B17/56** (20210101); **G03B17/02** (20210101); **G03B17/55** (20210101); **H04N23/53** (20230101)

U.S. Cl.:

CPC **G03B17/563** (20130101); **G03B17/02** (20130101); **G03B17/55** (20130101); **H04N23/531** (20230101);

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an image capturing apparatus.

Description of the Related Art

[0002] Conventionally, image capturing apparatuses, such as a camera and a video camera, include one to which a grip for being grasped by a user is attached. In recent years, there is a demand for size reduction of the grip. On the other hand, to improve the operability and the function of the image capturing apparatus, the grip is required to have a lot of operation keys arranged thereon. Even in a case where the grip is reduced in size and at the same time a lot of operation keys are arranged, there is a demand for ensuring the holdability and operability of the grip.

[0003] Japanese Patent No. 6492463 discloses a technique for connecting a grip to an image capturing apparatus body to which a plurality of lenses and a battery can be removably attached, via an arm, and changing the position of the grip with respect to the image capturing apparatus body. According to this technique, even when the center of gravity in an optical axis direction is changed due to a difference in weight between a lens and a battery, or the like, a user can stably perform image capturing regardless of a use form.

[0004] However, in the technique disclosed in Japanese Patent No. 6492463, the arm and the operation keys of the grip are arranged between the image capturing apparatus body and a grasping portion, which increases a distance between the image capturing apparatus body and the grasping portion. For this reason, it is difficult to reduce the size of the whole apparatus. Further, there is a problem that a load felt by a user with respect to a direction orthogonal to the optical axis is increased, which reduces the operability.

SUMMARY OF THE INVENTION

[0005] The present invention provides an image capturing apparatus that is improved in the operability while reducing the size of a grip.

[0006] The present invention provides an image capturing apparatus including an image capturing apparatus body; and a grip rotatably attached to a side surface of the image capturing apparatus body, wherein the grip includes a grasping portion for being grasped by a user, a first, a second, and a third operation elements for being operated by the user, a first exterior that covers the first operation element at least from the image capturing apparatus body, and a second exterior that covers the third operation element at least from the image capturing apparatus body, wherein out of rotational positions of the grip, when the grip is in a first rotational position in which the first operation element and the second operation element are positioned upward of the grasping portion, the third operation element is positioned between the first operation element and the second operation element at a location upward of a line segment connecting between the first operation element and the second operation element, and at the same time, the second exterior is more remote from the image capturing apparatus body than the first exterior.

[0007] According to the present invention, it is possible to improve the operability while reducing the size of the grip.

[0008] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a front perspective view of an image capturing apparatus according to a first embodiment.

[0010] FIG. **2** is a rear perspective view of the image capturing apparatus.

[0011] FIG. **3** is a rear perspective view of an image capturing apparatus body in a state in which a grip has been removed.

[0012] FIG. **4** is a front perspective view of the image capturing apparatus body in the state in which the grip has been removed.

[0013] FIG. **5** is a side view of the image capturing apparatus body to which the grip has been attached at a standard angle for use.

[0014] FIG. **6** is a rear view of the grip.

[0015] FIG. **7** is a rear view of the grip grasped by a user's right hand.

[0016] FIG. **8** is a cross-sectional view taken along MA**1**-MA**1** in FIG. **5**.

[0017] FIG. **9** is an enlarged view of a portion where operation keys are arranged.

[0018] FIG. **10** is a −X side view of the image capturing apparatus body to which the grip has been attached.

[0019] FIG. **11** is a +X side view of the grip.

[0020] FIG. **12** is a view of the image capturing apparatus body to which the grip has been attached, as viewed from a −Y side.

[0021] FIG. **13** is a front perspective view of an internal structure of the image capturing apparatus body.

[0022] FIG. **14** is a rear perspective view of the internal structure of the image capturing apparatus body.

[0023] FIG. **15** is a front exploded perspective view of the internal structure of the image capturing apparatus body.

[0024] FIG. **16** is a rear exploded perspective view of the internal structure of the image capturing apparatus body.

[0025] FIG. **17** is a top view of the internal structure.

[0026] FIG. **18**A is a rear view, and FIG. **18**B is a cross-sectional view, taken along NA-NA in FIG. **18**B, of the internal structure.

[0027] FIG. **19** is a top view of the image capturing apparatus body.

[0028] FIG. **20** is a left side view of the image capturing apparatus body.

[0029] FIG. **21** is a left side view of the image capturing apparatus at a time when a user grasps the grip.

[0030] FIGS. **22**A and **22**B are a front perspective view and a rear perspective view of the image capturing apparatus.

[0031] FIG. **23** is a front perspective view of the image capturing apparatus, from which illustration of a panel cable is omitted.

[0032] FIGS. **24**A and **24**B are an enlarged perspective view of the image capturing apparatus body and an enlarged perspective view of a panel unit.

[0033] FIG. **25** is a schematic view of the panel cable.

[0034] FIGS. **26**A and **26**B are a perspective view of a body-side retainer member and a perspective view of a panel-side retainer member.

[0035] FIGS. **27**A and **27**B are a rear perspective view and a front perspective view of part around a body-side insertion portion.

[0036] FIGS. **28**A and **28**B are an enlarged front view and an enlarged cross-sectional view, taken along OA-OA, of the part around the body-side insertion portion.

[0037] FIG. **29** is a front perspective view of the part around the body-side insertion portion at a time when the panel cable is not attached.
[0038] FIG. **30** is a front perspective view of an image capturing apparatus according to a second embodiment.
[0039] FIG. **31** is a front perspective view of the image capturing apparatus from which part of the component units has been separated.
[0040] FIGS. **32**A and **32**B are perspective views of the panel unit, as viewed from a display section side and a side opposite from the display section, respectively.
[0041] FIGS. **33**A to **33**D are perspective views of the panel unit.
[0042] FIG. **34** is a perspective view of a panel cable.
[0043] FIG. **35** is a perspective view of a panel cable.
[0044] FIGS. **36**A to **36**C are perspective views each showing a state in which a plug portion has been inserted in a socket portion.
[0045] FIGS. **37**A to **37**C are perspective views of the panel unit, and a view of the panel cable and a retainer member, as viewed from the inside.
[0046] FIGS. **38**A and **38**B are views of a panel unit of an image capturing apparatus according to a third embodiment, as viewed from a display section side and a side opposite from the display section, respectively.
[0047] FIG. **39** is a perspective view of a retainer member.
[0048] FIG. **40** is an exploded perspective view of part of the panel unit.
[0049] FIGS. **41**A to **41**F are perspective views and cross-sectional views of the panel unit.
[0050] FIGS. **42**A to **42**D are perspective views and a cross-sectional view of the panel unit, and a view of the retainer member and so forth, as viewed from the inside.
[0051] FIGS. **43**A to **43**D are perspective views and a cross-sectional view of the panel unit, and a view of the retainer member and so forth, as viewed from the inside.

DESCRIPTION OF THE EMBODIMENTS

[0052] The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.
[0053] FIGS. **1** and **2** are a front perspective view and a rea perspective view of an image capturing apparatus according to a first embodiment of the present invention.
[0054] Hereafter, directions of each component are referred to with reference to X, Y, and Z coordinate axes indicated in each drawing. Here, in a direction parallel to the center of an optical axis, an object side is referred to as the front (+Z direction) for convenience sake. The Z axis is perpendicular to an imaging surface of a sensor **101**. The Y axis is an axis representing a vertical direction of the image capturing apparatus, denoted by reference numeral **100**, and an upward direction is defined as a +Y direction. The X axis is an axis representing a lateral direction of the image capturing apparatus **100**, and a right direction as viewed from the object side is defined as a +X direction.
[0055] First, a configuration of the image capturing apparatus **100** will be described with reference to FIGS. **1** and **2**.
[0056] The image capturing apparatus **100** includes an image capturing apparatus body **104**, a handle **105**, a panel unit **106**, and a grip **111**. The image capturing apparatus body **104** has a lens attaching portion **102** and the sensor **101** that generates video data based on an optical image formed thereon, which are arranged on a front part. The lens attaching portion **102** is an attaching portion to which a various types of lenses which are different in optical performance can be removably attached. On a right side surface **103** (+X-side surface) of the image capturing apparatus **100**, operation members, such as a variety of buttons, are arranged. A user operates these operation members, whereby the user can perform a variety of operations, such as an operation of switching power on/off of the image capturing apparatus **100**, an image capturing operation for recording, and an audio adjustment operation.

[0057] On a top surface **109** (+Y-side surface) of the image capturing apparatus body **104**, the handle **105** used by a user to hold the image capturing apparatus **100** is provided. Further, the panel unit **106** for confirming an image capturing operation for recording is attached to the handle **105** via a panel unit-disposing mechanism **107**. With the panel unit-disposing mechanism **107**, it is possible to adjust the panel unit **106** to a desired position and to a desired angle with respect to the handle **105**.
[0058] The panel unit **106** is connected to the image capturing apparatus body **104** via a panel cable **108**. This enables transmission of a video signal, supply of electric power, and transmission and reception of a control signal, from the image capturing apparatus body **104** to the panel unit **106**. Details of how to attach the panel cable **108** to the image capturing apparatus body **104** and the panel unit **106** will be described hereinafter.
[0059] On a left side surface **110** (−X-side surface) of the image capturing apparatus body **104**, the grip **111** for holding the image capturing apparatus **100** is disposed. The grip **111** is attached in a state rotatable about the X axis with respect to the image capturing apparatus body **104**. The image capturing apparatus body **104** and the grip **111** are electrically connected to each other via a grip cable **113**. This enables supply of electric power, and transmission and reception of a control signal from the image capturing apparatus body **104** to the grip **111**. Details of the grip **111** will be described hereinafter.
[0060] The image capturing apparatus body **104** is equipped with a function of taking in outside air, exhausting heat generated from internal circuit boards to the outside together with exhaust wind, by using an air blower **360** (see FIG. **13**) to thereby keep the temperature of the image capturing apparatus body **104** at a temperature equal to or lower than a fixed temperature. To take in air, a first air inlet port **114** and a second air inlet port **115** are arranged in the left side surface **110** (−X-side surface) of the image capturing apparatus body **104**, and a third air inlet port **116** is arranged in the right side surface **103**. Further, an air exhaust port **117** for exhausting air warmed inside the image capturing apparatus body **104** is disposed in the left side surface **110**. Details of a heat dissipation structure will be described hereinafter.
[0061] Note that the image capturing apparatus **100** includes, in addition to these, a variety of components necessary to achieve a function for recording a moving image. However, details of these components and the function have less relation with the essence of the present invention, and hence description thereof is omitted.
[0062] Next, the details of the grip **111** will be described with reference to FIGS. **3** to **11**.
[0063] First, how to connect the image capturing apparatus body **104** and the grip **111** will be described with reference to FIGS. **3** and **4**.
[0064] FIGS. **3** and **4** are a rear perspective view and a front perspective view of the image capturing apparatus body **104** in a state in which the grip **111** has been removed, respectively. The grip **111** has a grip-side attaching portion **120** having rosette-shaped appearance, disposed on a grip rotational axis **122** indicated by a broken line. The grip-side attaching portion **120** has a male screw portion **124** which rotates interlockingly with a rotary operation portion **123**. The image capturing apparatus body **104** is provided with a body-side attaching portion **125** having rosette-shaped appearance. The body-side attaching portion **125** has a female screw portion **126** on the grip rotational axis **122**.
[0065] By aligning the respective centers of the grip-side attaching portion **120** and the body-side attaching portion **125** and rotating the rotary operation section **123**, the male screw portion **124** is screwed in the female screw portion **126**, whereby the grip **111** is attached to the image capturing apparatus body **104** (state shown in FIG. **2**). Note that the user can attach the grip **111** to the image capturing apparatus body **104** at a desired angle, with the grip rotational axis **122** in the center. With this, the user can perform shooting by setting the grip **111** at a preferred angle even when high-angle shooting or low-angle shooting is performed.
[0066] An end portion of the grip cable **113**, toward the grip **111**, is connected to an internal circuit

board of the grip **111** and extends from a wire connection position **127**, which is positioned at a location on the −Y side of the grip-side attaching portion **120**, in a direction between the −Z direction and the −Y direction. An end portion of the grip cable **113**, toward the image capturing apparatus body **104**, is comprised of a plug **128** and a housing portion **129** which is bent from the plug **128** in a direction substantially orthogonal thereto. A jack **131** which faces in a direction substantially orthogonal to the grip rotational axis **122** indicated by a broken line is provided at a location on the −Y side and the −Z side of the body-side attaching portion **125** of the image capturing apparatus body **104**. When the plug **128** is inserted into the jack **131**, the image capturing apparatus body **104** and the grip **111** are electrically connected to each other, thereby enabling supply of electric power and transmission and reception of a control signal.

[0067] FIG. **5** is a side view of the image capturing apparatus body **104** to which the grip has been attached at a standard angle for use. Part of the grip cable **113**, hidden by the grip **111**, is expressed by a broken line for convenience of explanation. As shown in FIG. **5**, the grip cable **113** has an excess length having an N-shape, and hence even when the grip **111** is rotated with respect to the image capturing apparatus body **104**, a level of tension which stretches the grip cable **113** to cause a break thereof is prevented from being applied to the grip cable **113**. Further, even if the grip cable **113** is pulled by an irregular motion, since the front end of the grip cable **113** has an L-shape, the plug **128** (see FIG. **3**) is difficult to be pulled off from the jack **131**. Further, even when the plug **128** is inserted into the jack **131** in a state in which the +Y-side end and the −Y-side end of the plug **128** are reversed, the plug **128** can be mechanically and electrically, for use.

[0068] Next, how to grasp the grip **111** and operate the rear-side keys will be described with reference to FIGS. **6** and **7**.

[0069] FIG. **6** is a rear view of the grip **111**. FIG. **7** is a rear view of the grip grasped by a user's right hand.

[0070] The grip **111** has a rear grasping portion **135**, a thumb placing portion **136**, and a hand back-fitting part **137** having upper and lower portions connected to the grip **111**. The rear grasping portion **135** is a portion for being grasped by a user. A first operation key **138**, a second operation key **139**, and a third operation key **140** are arranged at respective locations upward (+Y direction) of the rear grasping portion **135**. The operation keys **138**, **139**, and **140** are respective examples of a first operation element, a second operation element, and a third operation element, which are operated by the user.

[0071] In a case where a user grasps the grip **111**, the user grasps the rear grasping portion **135** with the base of the thumb, and the back side of the hand is brought into contact with the hand back-fitting part **137**. Further, the thumb is brought to a state in which the thumb is placed on the thumb placing portion **136**.

[0072] In a case where the user operates the first operation key **138**, the second operation key **139**, and the third operation key **140**, the user operates them using the thumb separated from the thumb placing portion **136**. Thus, since the operation keys are arranged at the respective locations upward (+Y side) of the rear grasping portion **135**, it is possible to make the position where the grip **111** is grasped closer to the image capturing apparatus body **104** than in a case where the operation keys are arranged on the left side (−X side) of the rear grasping portion **135**. This makes it possible to reduce a load applied to the user's hand by the weight of the image capturing apparatus **100**, which improves the operability.

[0073] FIG. **8** is a cross-sectional view taken along MAI-MAI in FIG. **5**. A positional relationship between the image capturing apparatus body **104** and the grip **111** and a positional relationship between the operation keys **138**, **139**, and **140** will be described with reference to FIG. **8**.

[0074] As shown in FIG. **8**, out of the rotational positions of the grip **111** with respect to the image capturing apparatus body **104**, a rotational position at which the operation keys **138**, **139**, and **140** are positioned upward of the rear grasping portion **135** is defined as a first rotational position.

[0075] A line segment connecting between the center C**1** of the first operation key **138** and the center C**2** of the second operation key **139** is defined as a line segment **145** indicated by a broken line. The center C**3** of the third operation key **140** is positioned upward of the line segment **145** and at the same time between the first operation key **138** and the second operation key **139**.

[0076] This key arrangement forming a triangle is advantageous for reducing the height of the third operation key **140** in the Y direction while separating the respective operation keys by a certain distance so as to prevent each operation key from being erroneously operated. By reducing the height of the third operation key **140** in the Y direction, the user can easily cause his/her thumb to reach the third operation key **140** even when the user has less freedom of his/her right hand due to the back side-fitting portion **137**, which can improve the operability. Further, by reducing the height of the third operation key **140** in the Y direction, it is possible to easily reduce the size of the whole grip **111** in the Y direction.

[0077] Further, the grip **111** has an upper grip exterior portion **146** (second exterior) and a lower grip exterior portion **148** (first exterior). The upper grip exterior portion **146** includes part covering the third operation key **140** at least from a side (+X side) of the image capturing apparatus body **104**. The lower grip exterior portion **148** includes part covering the first operation key **138** at least from a side (+X side) of the image capturing apparatus body **104**.

[0078] The image capturing apparatus body **104** is provided with an operation member **150** for operating the image capturing apparatus body **104**. The operation member **150** is disposed at a location opposed to the upper grip exterior portion **146** when the grip **111** is in the first rotational position.

[0079] With the above-described triangular key arrangement, the upper grip exterior portion **146** is more remote from the image capturing apparatus body **104** than the lower grip exterior portion **148** in the X direction. That is, it is possible to space the upper grip exterior portion **146** from the image capturing apparatus body **104** more than the lower grip exterior portion **148** by a distance **147**. Strictly describing, the position of the upper grip exterior portion **146** at the height of the center C**3** in the X direction is more remote from the capturing apparatus body **104** than the position of the lower grip exterior portion **148** at the height of the center C**1** in the X direction, by the distance **147**. With this, the user can operate the operation member **150** positioned at the location opposed to the upper grip exterior portion **146** with a sufficient allowance.

[0080] Further, when the grip **111** is in the first rotational position, the operation keys **138**, **139**, and **140** are positioned upward of the rear grasping portion **135** and do not overlap the rear grasping portion **135** in the X direction. This makes it easy to make the rear grasping portion **135** close to the image capturing apparatus body **104**, and hence this configuration is advantageous in reducing the size of the whole image capturing apparatus **100** in the X direction (width direction).

[0081] In the present embodiment, the first operation key **138** is a push-type key, the second operation key **139** is a cross key (direction input key), and the third operation key **140** is a push-type key. However, each of the operation keys **138**, **139**, and **140** can be any of a push-type key, a cross key, a slide key, a dial key, and a toggle key.

[0082] Note that although the line segment **145** is substantially parallel to the grip rotational axis **122** (see FIG. **3**), this is not limitative, but the line segment **145** can be at any other desired angle.

[0083] The operation member **150** is a member (such as a rotation lock member) for preventing the panel cable **108** from being pulled off from the image capturing apparatus body **104** when the panel cable **108** is inserted in the image capturing apparatus body **104**. However, from a limited viewpoint of capability of operating the operation member **150** with a sufficient allowance, the operation member **150** can be an operation section having another function.

[0084] FIG. **9** is an enlarged view of the portion where the operation keys **138**, **139**, and **140** are arranged. A distance relationship between the first operation key **138**, the second operation key **139**, and the third operation key **140** will be described with reference to FIG. **9**.

[0085] A distance between the center C**1** of the first operation key **138** and the center C**3** of the third operation key **140** (distance between the respective centers) is defined as a distance **157** (indicated by a broken line). A distance between the center C**2** of the second operation key **139** and the center C**1** of the first operation key **138** is defined as a distance **155** (indicated by a broken line). A distance between the center C**2** of the second operation key **139** and the center C**3** of the third operation key **140** is defined as a distance **156** (indicated by a broken line).

[0086] In general, in a case where a user operates a cross key in vertical and lateral directions, his/her finger largely moves in the vertical and lateral directions, compared with a case where the user operates a push-type key. Therefore, if another operation key is disposed in the vicinity of the cross key, there is large possibility that the user erroneously presses the other operation key.

[0087] To prevent this, the magnitude relation between the respective distances is set as the distance **157**<the distance **155**, and at the same time the distance **157**<the distance **156**. That is, the distances **155** and **156** between the cross key and the push-type keys are set to be longer than the distance **157** between the pus-type keys. With this, it is possible to prevent an erroneous operation, while reducing the size of the grip **111** by reducing the distances between the operation keys.

[0088] A positional relationship between the grip **111** and the air exhaust port **117** will be described with reference to FIGS. **10** to **12**.

[0089] FIG. **10** is a −X-side view of the image capturing apparatus body **104** to which the grip **111** has been attached. FIG. **11** is a +X-side view of the grip **111**. FIG. **12** is a view of the image capturing apparatus body **104** to which the grip **111** has been attached, as viewed from the −Y side (bottom view). In FIG. **10**, the outline of the air exhaust port **117**, hidden by the grip **111**, is indicated by a broken line for convenience of explanation. In FIG. **11**, a projected shape **161** of the air exhaust port **117** is indicated by a broken line.

[0090] Referring to FIGS. **10** to **12**, the grip **111** is in a second rotational position, which is different from the first rotational position, with respect to the image capturing apparatus body **104**. The second rotational position is a rotational position at which part of the air exhaust port **117** and part of the grip **111** overlap each other as viewed from an axial line direction of the grip rotational axis **122**, but the air exhaust port **117** does not overlap the lower grip exterior portion **148**. As shown in FIG. **11**, the upper grip exterior portion **146** overlaps the projected shadow of the air exhaust port **117**, in a state projected in the X direction, but the lower grip exterior portion **148** does not overlap the air exhaust port **117**.

[0091] As shown in FIG. **12**, the air exhaust port **117** of the image capturing apparatus body **104** protrudes toward the grip **111** (−X side) by a distance **160**, compared with the first air inlet port **114** and the second air inlet port **115**. However, only the upper grip exterior portion **146** of the grip **111** is opposed to the air exhaust port **117**, and hence it is possible to separate the air exhaust port **117** and the grip **111** by a distance indicated by an arrow **162** in the X direction, and heat discharge is not interfered with.

[0092] The internal structure of the image capturing apparatus body **104** will be described with reference to FIGS. **13** to **16**.

[0093] FIGS. **13**, **14**, **15**, and **16** are a front perspective view, a rear perspective view, a front exploded perspective view, and a rear exploded perspective view of the internal structure, denoted by reference numeral **300**, of the image capturing apparatus body **104**, respectively.

[0094] In the internal structure **300** of the image capturing apparatus body **104**, the sensor **101**, a sensor duct unit **320**, a main circuit board **330**, a main duct unit **340**, and a power supply board **350** are arranged substantially parallel from the front to the rear in the optical axis direction (+Z direction) in the mentioned order. Further, the air blower **360** and an exhaust duct unit **370**, a media duct unit **380**, a media board **390**, a sub media board **400**, and a wireless board **410** are arranged substantially parallel toward the rear in the mentioned order. A wireless antenna **420** is disposed in upper part (+Y side) of rear part (−Z side) of the internal structure **300**.

[0095] The sensor **101** converts light incident through a lens, not shown, to electrical signals. The main circuit board **330** is a board that controls the overall operation of the image capturing apparatus **100** and is the largest in area in the image capturing apparatus **100** in which a lot of integrated circuits (ICs) are mounted. On the main circuit board **330**, an IC for processing a signal output from the sensor **101**, an IC for performing processing, such as processing for compressing a video, a memory used by these ICs, and so forth, are mounted.

[0096] The power supply board **350** supplies electric power to all the electric devices in the image capturing apparatus **100**, including the above-mentioned main circuit board **330**. The media board **390** is a board for recording a video in a recording medium. The sub media board **400** is a board for storing settings used at the time of shooting, and recording a video for backup, which is reduced in data volume of a video to be recorded by the media board **390**. The wireless board **410** is electrically connected to the wireless antenna **420** via a wire **430** (see FIG. **17**) to control wireless communication with an external apparatus.

[0097] The heat dissipation structure of the image capturing apparatus body **104** will be described with reference to FIGS. **17**, **18**A, and **18**B.

[0098] FIG. **17** is a top view of the internal structure **300**. FIG. **18**A is a rear view of the internal structure **300**. FIG. **18**B is a cross-sectional view taken along NA-NA in FIG. **18**A.

[0099] The image capturing apparatus **100** dissipates heat from a variety of heat sources by forced air cooling using the heat dissipation ducts and the air blower **360**. The heat dissipation ducts are formed by the above-mentioned sensor duct unit **320**, the main duct unit **340**, the media duct unit **380**, and the exhaust duct unit **370**.

[0100] The sensor duct unit **320** takes in outside air from sensor duct air inlet ports **321**, and the taken-in air flows from a sensor duct connection portion **342** (see FIG. **15**) of the main duct unit **340** into the main duct unit **340**. The media duct unit **380** takes in outside air from a media duct air inlet port **381**, and the taken-in air flows from a main duct connection portion **343** (see FIG. **16**) of the main duct unit **340** into the main duct unit **340**. The main duct unit **340** takes in outside air from a main duct air inlet port **341**, and the taken-in air flows into a blower connection portion **344** (see FIG. **16**).

[0101] The air blower **360** takes in air from the blower connection portion **344** of the main duct unit **340**. Therefore, the air flowing from the sensor duct unit **320** and the media duct unit **380** into the main duct unit **340** and the air taken in from the main duct air inlet port **341** into the main duct unit **340** merge in the main duct unit **340**. Then, the merged air is taken in by the air blower **360** through the blower connection portion **344**. The taken-in air is discharged out of the image capturing apparatus **100** through the exhaust duct unit **370**.

[0102] Here, the sensor duct air inlet ports **321** and the first air inlet port **114** (see FIG. **2**) are connected, and the main duct air inlet port **341** and the second air inlet port **115** (see FIG. **2**) are connected. Further, the media duct air inlet port **381** and the third air inlet port **116** (see FIG. **1**) are connected, and the exhaust duct unit **370** and the air exhaust port **117** (see FIG. **2**) are connected. The first air inlet port **114** and the second air inlet port **115**, and the sensor duct air inlet ports **321** and the main duct air inlet port **341** (see FIG. **18**B) are positioned more toward the object side (+Z side) in the optical axis direction than the air exhaust port **117**.

[0103] The above-mentioned air flow passage configuration of the forced air cooling in the image capturing apparatus **100** is illustrated by broken lines in FIGS. **17**, **18**A, and **18**B.

[0104] Heat generated in the sensor **101** is transferred to the sensor duct unit **320** via a heat-conductive member, such as a graphite sheet, not shown. Heat generated on a +Z-side surface of the main circuit board **330** is transferred to the sensor duct unit **320** via a heat-conductive member, such as heat dissipation rubber, not shown, and heat generated on a −Z-side surface of the main circuit board **330** is transferred to the main duct unit **340** via a heat-conductive member, such as heat dissipation rubber, not shown. Heat generated in the power supply board **350** is transferred to the main duct unit **340** via a heat-conductive member, such as heat dissipation rubber, not shown.

[0105] The media board **390** is mounted in an opening **382** (see FIGS. **16** and **18**B) of the media duct unit **380**, whereby at least part of the media board **390** is exposed in the media duct unit **380**. Therefore, air flowing in the media duct unit **380** is in direct contact with the media board **390**, and hence a recording medium inserted in the media board **390** is cooled by dissipating heat from the media board **390** itself. By efficiently cooling the recording medium, it is possible to stably perform recording even when power consumption of the recording medium is increased by high-bit rate data e.g. due to high resolution of recent videos.

[0106] The sub media board **400** is arranged substantially parallel to the media board **390** at a location rearward (on the-Z side) of the media board **390**. The sub media board **400** performs writing of data which is small in data volume, as described above, and hence the amount of generated heat is also small, so that heat dissipation using forced air cooling is not necessarily required.

[0107] The wireless board **410** is disposed on substantially the same plane as the sub media board **400**, and at the same time, on the-X side of the sub media board **400** (see FIG. **18**B). Heat generated in the wireless board **410** is transferred to the media duct unit **380** via a heat-conductive member, not shown. A wireless control IC, not shown, mounted on the wireless board **410** is lower in guaranteed temperature than the other heat sources (the sensor **101**, the main circuit board **330**, the power supply board **350**, and the media board **390**) of the internal structure **300**. So, by mounting the wireless control IC not on the main circuit board **330**, but on the wireless board **410** as an independent board, it is possible to dissipate heat without being affected by the other heat sources having high power consumption.

[0108] With the above-described heat dissipation structure of the image capturing apparatus body **104**, it is possible to transfer heat of the variety of heat sources of the internal structure **300** to the variety of heat dissipation ducts, which are opposed and adjacent to the heat sources, respectively, exchange the heat to air by the above-described flow passage configuration, and efficiently discharge the heat to the outside of the image capturing apparatus body **104**.

[0109] The arrangement of the air exhaust port **117** of the image capturing apparatus body **104** will be described with reference to FIGS. **19** to **21**.

[0110] FIGS. **19** and **20** are a top view and a left side view of the image capturing apparatus body **104**. FIG. **21** is a left side view of the image capturing apparatus **100** at a time when the user grasps the grip **111**.

[0111] The air exhaust port **117** is disposed at an end portion of the left side surface **110** on an opposite side (−Z side) thereof to the object side in the optical axis direction, and at the same time at a location upward of the optical axis **130**. The air exhaust port **117** opens in a direction substantially perpendicular to the left side surface **110** (toward the −X side).

[0112] Similarly, the first air inlet port **114** and the second air inlet port **115** also open toward the-X side substantially perpendicularly to the left side surface **110** and are adjacent to the air exhaust port **117**, and hence if no measure is employed, warmed exhaust air wind can be taken in. However, as shown in FIG. **17**, wind discharged from the air blower **360** is discharged obliquely rearward by the exhaust duct unit **370**. This prevents the first air inlet port **114** and the second air inlet port **115** from taking in the exhaust air.

[0113] As indicated by a broken line in FIG. **19**, the body-side attaching portion **125** more protrudes outward of the image capturing apparatus body **104** in a direction (X direction) perpendicular to the left side surface **110** than the air exhaust port **117**. That is, as viewed from the top surface of the image capturing apparatus body **104**, the air exhaust port **117** is accommodated more toward the +X side than the body-side attaching portion **125**. This is advantageous in reducing the size of the image capturing apparatus body **104** in the width direction (X direction).

[0114] Further, as shown in FIG. **20**, the air exhaust port **117** is disposed in

[0115] the left side surface **110**, at a location opposite from the object in the optical axis direction, and at the same time upward of the body-side attaching portion **125**. With this, as shown in FIG. **21**, when a photographer grasps the grip **111** during shooting performed by using the image capturing apparatus **100**, a hand **440** is prevented from blocking the air exhaust port **117**. If the air exhaust port **117** is blocked during shooting, there is a possibility that air cannot be sufficiently discharged, resulting in lowered heat dissipation performance. Further, if the warmed exhaust air is blown to the photographer, uncomfortable feeling is given to the photographer. This situation is prevented.

[0116] With this arrangement of the air exhaust port **117**, the air exhaust port **117** is not blocked during shooting, and the exhaust wind is not blown to the photographer, and hence it is possible to suppress lowering of the heat dissipation performance and uncomfortable feeling given to the photographer.

[0117] The configuration of the panel unit **106** will be mainly described with reference to FIGS. **22**A to **24**B.

[0118] FIGS. **22**A and **22**B are a front perspective view and a rear perspective view of the image capturing apparatus **100**, respectively. FIG. **23** is a front perspective view of the image capturing apparatus **100**, from which illustration of the panel cable **108** is omitted. The panel unit **106** is a display device communicably connected to the image capturing apparatus body **104** by the panel cable **108**.

[0119] The image capturing apparatus **100** is configured to convert light captured by the sensor **101** to signals and send the signals to the display section **511** of the panel unit **106** via the panel cable **108**. Note that detailed description of a process for performing this shooting is not related to the present invention, and hence the description thereof is omitted.

[0120] First, details of how to attach the panel cable **108** connecting between the image capturing apparatus body **104** and the panel unit **106** will be described.

[0121] As shown in FIG. **23**, in the image capturing apparatus body **104**, a body-side insertion portion **551** in which the panel cable **108** is inserted is disposed in an inclined surface **562** at a location toward the −X side and at the same time on the +Y side of the image capturing apparatus body **104**. By arranging the body-side insertion portion **551** on the inclined surface **562** of the image capturing apparatus body **104**, it is possible to provide the body-side insertion portion **551** in a space between the image capturing apparatus body **104** and the grip **111**, and hence prevent increasing the size of the image capturing apparatus body **104**. Further, on the panel unit **106**, a panel-side insertion portion **552** in which the panel cable **108** is inserted is disposed on a back surface (rear surface) opposite from the display section **511**.

[0122] Here, a peripheral structure of the body-side insertion portion **551** and the panel-side insertion portion **552** will be described with reference to FIGS. **24**A and **24**B. FIG. **24**A is an enlarged perspective view of the image capturing apparatus body **104**, which illustrates the peripheral structure of the body-side insertion portion **551**. FIG. **24**B is an enlarged perspective view of the panel unit **106**, which illustrates the peripheral structure of t the panel-side insertion portion **552**.

[0123] First, the body-side insertion portion **551** will be described. As shown in FIG. **24**A, on the image capturing apparatus body **104**, a body-side socket portion **561** (first socket portion) to which the panel cable **108** is removably inserted is disposed such that its insertion direction is substantially parallel to the inclined surface **562** of the image capturing apparatus body **104**. Further, an opening of the body-side socket portion **561** (insertion/removal opening into which the panel cable **108** is removably inserted) opens toward the object side. An insertion/removal direction **563** of the panel cable **108** with respect to the image capturing apparatus body **104** is indicated by a broken-line arrow in FIG. **24**A. The panel cable **108** is inserted into the body-side socket portion **561** from the +Z direction to the −Z direction. The insertion/removal direction **563** is substantially parallel to the optical axis.

[0124] The image capturing apparatus body **104** has a body-side shaft **571** disposed along the insertion/removal direction **563** and a body-side retainer member **572** supported in a state rotatable about the body-side shaft **571** (about a first axial line) within a predetermined range. The axial line of the body-side shaft **571** is substantially parallel to the insertion/removal direction **563**.

[0125] The body-side retainer member **572** has a hole **576**. The operation member **150** is inserted

through the hole **576** and is integrated with the body-side retainer member **572** by a screw retainer member **574**.
[0126] The image capturing apparatus body **104** has a female screw portion **577** substantially orthogonal to the insertion/removal direction **563**. When the operation member **150** is screwed into the female screw portion **577**, the body-side retainer member **572** is fixed. Further, the inclined surface **562** is formed with a recess shaped portion **601** for avoiding the outer shape of the panel cable **108** and a rib **611** for receiving static pressure of the panel cable **108**. Details of the recess shaped portion **601** and the rib **611** will be described hereinafter.
[0127] Here, an outer angular shape of the image capturing apparatus body **104** has a substantially rectangular shape, as viewed from the object side, and the body-side socket portion **561** and the body-side retainer member **572** are arranged in one corner (vertex on the −X side and at the same time on the +Y side) of the rectangular shape.
[0128] Next, the panel-side insertion portion **552** will be described. As shown in FIG. **24**B, on the panel unit **106**, a panel-side socket portion **581** (second socket portion) for removably inserting the panel cable **108** therein is disposed on a surface (rear surface) of the panel unit **106**, which is opposite from the display section **511**. The panel-side socket portion **581** is disposed substantially parallel to the display surface of the display section **511**.
[0129] Further, when the panel unit **106** is in a posture shown in FIG. **22**A,
[0130] an opening of the panel-side socket portion **581** (insertion/removal opening into which the panel cable **108** is removably inserted) opens toward the −X side. An insertion/removal direction **580** of the panel cable **108** into and from the panel unit **106** is indicated by a broken-line arrow in FIG. **24**B. When the panel unit **106** is in the posture shown in FIG. **22**A, the panel cable **108** is inserted into the panel-side socket portion **581** approximately from the −X direction toward the +X direction.
[0131] Further, when the panel unit **106** is in the posture shown in FIG. **22**A, the opening direction of the insertion/removal opening can be tilted by 6 degrees in the +Z direction with respect to the display surface of the display section **511**. By tilting the insertion/removal opening as described above, the user can easily removably insert the panel cable **108**. The insertion/removal direction **580** is substantially parallel to the rear surface of the panel unit **106**.
[0132] The panel unit **106** has a panel-side shaft **582** and a panel-side retainer member **583**. The panel-side shaft **582** is disposed substantially parallel to the display section **511** along the insertion/removal direction **580** of the panel cable **108**. The panel-side retainer member **583** is supported in a state rotatable about the panel-side shaft **582** within a predetermined range. The axial line (second axial line) of the panel-side shaft **582** is substantially parallel to the insertion/removal direction **580**.
[0133] The panel-side retainer member **583** has a hole **584** (see FIG. **26**B). A display section-side screw member **585** is inserted through the hole **584** and is integrated with the panel-side retainer member **583** by a screw retainer member **586**. The panel unit **106** has a display section-side female screw portion **587** substantially orthogonal to the insertion/removal direction **580**. When the display section-side screw member **585** is screwed into the display section-side female screw portion **587**, the panel-side retainer member **583** is fixed.
[0134] The configuration of the panel cable **108** and a retaining structure of the panel cable **108** will be described with reference to FIG. **25**.
[0135] FIG. **25** is a schematic view of the panel cable **108**. The panel cable **108** includes plug portions **599** which can be inserted and fitted in both the body-side socket portion **561** and the panel-side socket portion **581**, respectively.
[0136] The panel cable **108** further includes grasping portions **590** each for being held by the user when the panel cable **108** is inserted or removed and a bending portion **591** as a flexible member which is flexibly bent.
[0137] Note that the panel cable **108** has the plug portion **599**, the grasping portion **590**, the

bending portion **591**, the grasping portion **590**, and the plug portion **599**, arranged in the mentioned order from one end to the other end thereof. Further, each grasping portion **591** is formed with protruding portions **592**. Therefore, the panel cable **108** has the two grasping portions **590** and the bending portion **591** as the flexible member that connects between the two grasping portions **591**, and the protruding portions **592** are disposed on opposite ends of the bending portion **591**.

[0138] The protruding portions **592** are disposed on both of the front and back sides of each grasping portion **590** (both side surfaces of each grasping portion **590** in a direction perpendicular to the insertion/removal direction of each plug portion **599**) but can be arranged only on one side. Each protruding portion **592** is provided such that it protrudes in a direction substantially orthogonal to an insertion/removal direction of the plug portion **599**. A surface of each protruding portion **592**, which is substantially perpendicular to the insertion/removal direction of the panel cable **108** and toward the bending portion **591**, is a cable engaging surface **593**. The cable engaging surface **593** plays a role of retaining the panel cable **108**. The retaining function of the cable engaging surface **593** will be described hereinafter.

[0139] FIGS. **26**A and **26**B are a perspective view of the body-side retainer member **572** and a perspective view of the panel-side retainer member **583**. The body-side retainer member **572** is formed with a body-side recess portion **594** and a body-side retainer engaging surface **595** (first engaging portion). The panel-side retainer member **583** is formed with a panel-side recess portion **596** and a panel-side retainer engaging surface **597** (second engaging portion). The body-side recess portion **594** and the panel-side recess portion **596** are avoiding portions for avoiding the protruding portions **592** (see FIG. **25**) of the panel cable **108**. The body-side retainer engaging surface **595** and the panel-side retainer engaging surface **597** each are opposed to the cable engagement surface **593** (see FIG. **25**) of the panel cable **108** to be inserted.

[0140] FIGS. **27**A and **27**B are a rear perspective view and a front erspective view of part around the body-side insertion portion **551** of the image capturing apparatus body **104** to which the panel cable **108** is attached, respectively. A structure for retaining the panel cable **108** by the body-side retainer member **572** will be described.

[0141] After the plug portion **599** (see FIG. **25**) is fitted in the body-side socket portion **561**, when the operation member **150** of the body-side retainer member **572** is screwed into the female screw portion **577**, the cable engagement surface **593** of the protruding portion **592** and the body-side retainer engaging surface **595** are opposed in the insertion/removal direction **563** and engaged with each other. With this, the movement of the panel cable **108** along the insertion/removal direction **563** is restricted, whereby it is possible to prevent the panel cable **108** from being carelessly pulled off when the user uses the image capturing apparatus body **104**. That is, when the plug portion **599** is inserted into the socket portion, the body-side retainer engaging surface **595** is engaged with the protruding portion **592**, whereby the function of retaining the plug portion **599** by the body-side retainer member **572** as the first retainer member is achieved.

[0142] FIG. **28**A is an enlarged front view of the part around the body-side insertion portion **551** at a time when the image capturing apparatus body **104** to which the panel unit **106** has been attached is viewed from the object side (+Z direction). FIG. **28**B is an enlarged cross-sectional view of the part around the body-side insertion portion **551**, taken along OA-OA in FIG. **28**A. Note that FIGS. **28**A and **28**B each show a state in which the panel cable **108** is attached and the operation member **150** is screwed in the female screw portion **577**.

[0143] The image capturing apparatus body **104** is formed with a recess shaped portion **601** for avoiding the protruding portion **592** when the plug portion **599** is inserted into the body-side socket portion **561**. In the direction of insertion/removal of the plug portion **599**, a length a of the recess shaped portion **601** is longer than a length b of engagement between the plug portion **599** of the panel cable **108** and the body-side socket portion **561**. This prevents insertion and removal of the panel cable **108** from being interfered with.

[0144] Further, an end portion of the recess shaped portion **601** is formed with a rib **611** having a protruding shape. The rib **611** is brought into contact with the grasping portion **590** when the plug portion **599** is inserted into the body-side socket portion **561** and urged from the +Y direction toward the-Y direction. With this, the rib **611** supports the panel cable **108**, and it is possible to prevent the plug portion **599** and the body-side socket portion **561** from being damaged.

[0145] Note that the above description has been given of the structure for retaining the panel cable **108** by using the body-side retainer member **572** in the image capturing apparatus body **104**. Although detailed description is omitted, similar to the image capturing apparatus body **104**, the panel-side retainer member **583** in the panel unit **106** also realizes the retaining structure by engagement between the cable engagement surface **593** of the protruding portion **592** and the panel-side engagement surface **597** (see FIG. **26**B). That is, when the plug portion **599** is inserted into the panel-side socket portion **581** (see FIG. **24**B), the panel-side engagement surface **597** is engaged with the protruding portion **592**, whereby the function of retaining the plug portion **599** by the panel-side retainer member **583** as the second retainer member is achieved.

[0146] Further, although not shown, the panel unit **106** can be formed with a recess shaped portion corresponding to the recess shaped portion **601** for avoiding the protruding portion **592** when the plug portion **599** is inserted into the panel-side socket portion **581**. Further, in the direction of insertion and removal of the plug portion **599**, a length of the above-mentioned recess shaped portion can be set to be longer than a length of engagement between the plug portion **599** and the panel-side socket portion **581**. Further, the panel unit **106** can be formed with a protruding shape portion corresponding to the rib **611**, which is brought into contact with the grasping portion **590** when the plug portion **599** is inserted into the panel-side socket portion **581**.

[0147] Note that the above-described retaining structure can be applied to only one of the image capturing apparatus body **104** and the panel unit **106**.

[0148] FIG. **29** is a front perspective view of the part around the body-side insertion portion **551** at a time when the panel cable **108** is not attached. A configuration of the image capturing apparatus body **104** at a time when the panel cable **108** is not attached will be described with reference to FIG. **29**.

[0149] When the panel cable **108** is not attached, the body-side socket portion **561** (see FIG. **24**A) can be protected by a protection cover **621** as a cover portion which can be engageable with the body-side socket portion **561**. The protection cover **621** is attached to the image capturing apparatus body **104** by a screw **622** and is rotated about the screw **622**. The outer shape of the protection cover **621** is a shape for being fitted in the body-side recess portion **594** of the body-side retainer member **572**. In the state shown in FIG. **29**, when the operation member **150** of the body-side retainer member **572** is screwed into the female screw portion **577**, the body-side retainer member **572** plays a role of restricting the movement of the protection cover **621**. That is, since the body-side retainer member **572** is fixed to the image capturing apparatus body **104** in the state in which the protection cover **621** is engaged with the body-side socket portion **561**, the movement of the protection cover **621** is restricted. Note that this configuration can also be applied to the panel-side insertion portion **552**.

[0150] According to the present embodiment, with the above-described triangular arrangement of the operation keys **138**, **139**, and **140** and the arrangement of the upper grip exterior portions **146** and the lower grip exterior portion **148**, it is possible to improve the operability while reducing the size of the grip **111**.

[0151] Next, a second embodiment of the present invention will be described. FIG. **30** is a front perspective view of an image capturing apparatus according to the second embodiment. This image capturing apparatus, denoted by reference numeral **700**, includes the image capturing apparatus body **104**, a lens barrel **701**, a panel unit **702**, and a panel cable **703**. The configuration of the image capturing apparatus body **104** is the same as that of the first embodiment.

[0152] The panel unit **702** can take a desired posture with respect to the image capturing apparatus body **104** according to rotational operations about a plurality of rotational shafts **705** provided in a hinge unit **704** and a sliding operation performed by using a slider **706**. The panel unit **702** can receive video signals from the image capturing apparatus body **104** via the panel cable **703** and display a video on a display section **707** (see FIG. **32**A), described hereinafter, after processing the video signals by an internal processor (not shown).

[0153] FIG. **31** is a front perspective view of the image capturing apparatus **700** from which part of the component units has been separated. The panel cable **703** can be removably attached to the body-side socket portion **561** of the image capturing apparatus body **104** and a socket portion **711** of the panel unit **702**, respectively. That is, a plug portion **710** of the panel cable **703** is held in a state insertable into and removable from the body-side socket portion **561** by a known configuration. Similarly, the plug portion **710** is held in a state insertable into and removable from the socket portion **711** by a known configuration. This enables the image capturing apparatus body **104** and the panel unit **702** to communicate with each other.

[0154] In the present embodiment, the plug portion **710** and the socket portion **711** conform to the USB Type-C standard. The panel unit **702** has a female screw portion **712** and can be removably attached to the hinge unit **704** by a male screw member **714**.

[0155] FIGS. **32**A and **32**B are perspective views of the panel unit **702**, as viewed from the side of the display section **707** and a side opposite from the display section **707**, respectively. In FIG. **32**B, part of the panel cable **703** is additionally illustrated together with the panel unit **702** for convenience of explanation.

[0156] The panel unit **702** has a substantially rectangular parallelepiped shape and has a main surface **713** on which the display section **707** is provided. The socket portion **711** is exposed from a surface **715** (rear surface) opposite from the main surface **713**. The panel unit **702** has a shaft **716** and a retainer member **717**. The shaft **716** is disposed along (substantially parallel to) an insertion/removal direction SA of the panel cable **703**. The retainer member **717** is supported in a state slidable along the shaft **716** within a predetermined range and at the same time rotatable about the shaft **716** within a predetermined range. The retainer member **717** has a slit hole **718** formed along the insertion/removal direction SA. A screw member **719** is inserted through the slit hole **718**. The retainer member **717** is slidable along the insertion/removal direction SA.

[0157] The panel unit **702** has a female screw portion **720** arranged to be substantially orthogonal to the insertion/removal direction SA. By screwing the screw member **719** as a fixing member into the female screw portion **720**, the retainer member **717** can be fixed to a desired position along the shaft **716** within a predetermined range. In the present embodiment, a state in which the retainer member **717** is fixed is referred to as the “closed state”, and an opened state in which the retainer member **717** does not overlap the socket portion **711** as viewed from the axial direction of the shaft **716** is referred to as the “retreated state”.

[0158] FIGS. **33**A and **33**C are perspective views of the panel unit **702** in a state in which the retainer member **717** is in the retreated state. FIGS. **33**B and **33**D are perspective views of the panel unit **702** in a state in which the retainer member **717** is in the closed state. FIGS. **33**A and **33**B each show a state in which the retainer member **717** is made close to the socket portion **711** along the shaft **716**, and FIGS. **33**C and **33**D each show a state in which the retainer member **717** is made remote from the socket portion **711** along the shaft **716**.

[0159] A position in which the retainer member **717** is in the closed state is referred to as the first position (see e.g. FIG. **33**B), and a position in which the retainer member **717** is in the retreated state is referred to as the third position (see e.g. FIG. **33**A). As for the sliding position in the insertion/removal direction SA, the retainer member **717** is slidable between the first position close to the socket portion **711** and a second position (see e.g. FIG. **33**D) remote from the socket portion **711**. A position where a cable engagement surface **726** (FIG. **36**C) retreats from between the first position and the second position by rotating the retainer member **717** about the axial center of the shaft **716** is the third position. When the retainer member **717** is rotated to the third position, the plug portion **710** can be inserted into and removed from the socket portion **711**.

[0160] Next, a plurality of panel cables will be described with reference to FIGS. **34** and **35**. In the image capturing apparatus **700** according to the present embodiment, the panel cable **703** and a panel cable **723**, which are different in the shape of the grasping portion from each other, can be selectively used.
[0161] FIG. **34** is a perspective view of the panel cable **703**. The panel cable **703** has a bending portion **721** as a flexible member which is flexibly bent, the plug portion **710**, and a grasping portion **722** (cable grasping portion). The grasping portion **722** is formed with a retainer member contact surface **728**.
[0162] FIG. **35** is a perspective view of the panel cable **723**. The panel cable **723** has the bending portion **721** as a flexible member which is flexibly bent, the plug portion **710**, and a grasping portion **725**. The grasping portion **725** is formed with a retainer member contact surface **729**. The panel cable **703** and the panel cable **723** are different in the length of the grasping portion **722** and the length of the grasping portion **725** in the plug insertion/removal direction SA.
[0163] FIGS. **36**A and **36**B are perspective views each showing a state in which the plug portion **710** (see FIG. **34**) of the panel cable **703** is inserted in the socket portion **711** of the panel unit **702**. In FIGS. **36**A and **36**B, the retainer member **717** is in the position close to the socket portion **711** along the insertion/removal direction SA (the same position as shown in FIGS. **33**A and **33**B). That is, FIGS. **36**A and **36**B show the retainer member **717** in the retreated state and the closed state, respectively.
[0164] FIG. **36**C is a view of the panel cable **703** and the retainer member **717** at a time when the retainer member **717** is in the closed state, as viewed from the inside of the retainer member **717**. The retainer member **717** has the cable engagement surface **726** as an engagement portion orthogonal to the insertion/removal direction SA. As shown in FIG. **36**C, the cable engagement surface **726** of the retainer member **717** in the closed state is in contact with the retainer member contact surface **728** of the grasping portion **722** of the panel cable **703** to prevent the plug portion **710** from being pulled off from the socket portion **711**.
[0165] FIGS. **37**A and **37**B are perspective views each showing a state in which the plug portion **710** (see FIG. **35**) of the panel cable **723** is inserted in the socket portion **711** of the panel unit **702**. In FIGS. **37**A and **37**B, the retainer member **717** is in the position remote from the socket portion **711** along the insertion/removal direction SA (the same position as shown in FIGS. **33**C and **33**D). That is, FIGS. **37**A and **37**B show the retainer member **717** in the retreated state and the closed state, respectively.
[0166] FIG. **37**C is a view of the panel cable **723** and the retainer member **717** at a time when the retainer member **717** is in the closed state, as viewed from the inside of the retainer member **717**. As shown in FIG. **37**C, the cable engagement surface **726** of the retainer member **717** in the closed state is in contact with the retainer member contact surface **729** of the grasping portion **725** of the panel cable **723** to prevent the plug portion **710** from being pulled off from the socket portion **711**.
[0167] Thus, when the plug portion **710** is inserted in the socket portion **711**, the cable engagement surface **726** is engaged with the grasping portion (**722** or **725**) in the first position or the second position, whereby the function of retaining the plug portion **710** by the retainer member **717** is achieved.
[0168] Further, when the screw member **719** and the female screw portion **720** are screwed with each other, the retainer member **717** is fixed to the body of the panel unit **702**. At this time, the screw member **719** urges part of the retainer member **717** toward the body of the panel unit **702**. With this, the sliding movement of the retainer member **717** is restricted.
[0169] According to the present embodiment, it is possible to obtain the same advantageous effects as provided by the first embodiment with respect to improvement of the operability while reducing the size of the grip **111**.
[0170] Further, the retainer member **717** can take the first position and the second position. Therefore, even when the panel cables (**703** and **723**) which are different in the length of the

grasping portion (**722** and **725**) are selectively used, it is possible to prevent the plug portion of the panel cable from being pulled off from the socket portion **711**.
[0171] Next, a third embodiment of the present invention will be described. The third embodiment differs from the second embodiment in the structure for retaining the panel cable and is the same as the second embodiment in the other components.
[0172] FIGS. **38**A and **38**B are views of a panel unit **801** of an image capturing apparatus according to the third embodiment, as viewed from a side of a display section **803** and a side opposite from the display section **803**, respectively. In FIG. **38**B, part of the panel cable **703** is additionally illustrated together with the panel unit **801** for convenience of explanation.
[0173] The panel unit **801** has a substantially rectangular parallelepiped shape and has a main surface **802** on which the display section **803** is provided. A socket portion **805** is exposed from a surface **804** (rear surface) opposite from the main surface **802**. The panel unit **801** has a retainer member **806**. The retainer member **806** is slidable along the insertion/removal direction SA of the panel cable **703** within a predetermined range. Further, the retainer member **806** is rotatable about an axial center **807** substantially orthogonal to the insertion/removal direction SA (slide direction) within a predetermined range. An axial line of stoppers **810**, described hereinafter, is the axial center **807** (see FIG. **39**).
[0174] FIG. **39** is a perspective view of the retainer member **806**. The retainer member **806** has a slide portion **809**, the stoppers **810**, and a cable engagement surface **811**. The slide portion **809** has a slit **808**. The axial center **807** of the stoppers **810** is orthogonal to a longitudinal direction SD of the slit **808**. The cable engagement surface **811** as the engagement portion is provided on an end of the retainer member **806**, which is opposite from the stoppers **810**, such that it is substantially orthogonal to the longitudinal direction SD of the slit **808**.
[0175] FIG. **40** is an exploded perspective view of part of the panel unit **801**. The panel unit **801** has an accommodating portion **812** for accommodating the slide portion **809**, a slit **813** in which the stoppers **810** can slide, and a female screw portion **814**.
[0176] As described hereinafter, the retainer member **806** can be pulled out up to a position where the stoppers **810** are brought into abutment with an end **815** of the slit **813**. In the position where the stoppers **810** are in contact with the end **815** of the slit **813**, the retainer member **806** is rotatable about the axial center **807** of the stoppers **810**. A screw member **816** as a fixing member is inserted through the slit **808** and screwed into the female screw portion **814**.
[0177] FIGS. **41**A, **41**C, and **41**E are perspective views of the panel unit **801**. FIGS. **41**B, **41**D, and **41**F are cross-sectional views each taken along a plane vertically extending through the screw member **816**, which passes the socket portion **805** and the screw member **816**.
[0178] The retainer member **806** can take a first position shown in FIGS. **41**A and **41**B, a second position shown in FIGS. **41**C and **41**D, and a third position shown in FIGS. **41**E and **41**F.
[0179] When the retainer member **806** is in the first position (FIGS. **41**A and **41**B), the slide portion **809** is in a state disposed in the accommodating portion **812**. In the first position, by screwing the screw member **816** into the female screw portion (see FIG. **40**), the retainer member **806** can be fixed. The retainer member **806** urges the slide portion **809** toward the body of the panel unit **801**.
[0180] When the retainer member **806** is in the second position (FIGS. **41**C and **41**D), the retainer member **806** is in a state pulled out to the position where the stoppers **810** are brought into abutment with the end **815** of the slit **813**. In the second position, it is also possible to fix the retainer member **806** by the screw member **816**, and the retainer member **806** urges the slide portion **809** toward the body of the panel unit **801**.
[0181] That is, when the retainer member **806** is in the first position or the second position, the retainer member **806** is fixed to the body of the panel unit **801** by screwing between the screw member **816** and the female screw portion **814**. At this time, the screw member **816** urges part of the retainer member **806** toward the body of the panel unit **801**. With this, the sliding movement of

the retainer member **806** is restricted.

[0182] The third position (FIGS. **41**E and **41**F) shows a state in which the retainer member **806** is rotated about the stoppers **810** from the second position.

[0183] When the retainer member **806** is in the third position, by loosening the screw member **816**, it is possible to release the slide portion **809** from the urged state and rotate the retainer member **806** to the second position.

[0184] As for the sliding position in the insertion/removal direction SA, the retainer member **806** can slide between the first position close to the socket portion **805** and the second position remote from the socket portion **805**. A position where the retainer member **806** is rotated about the axial center **807**, thereby causing the cable engagement surface **811** (see FIG. **40**) to retreat from between the first position and the second position, is the third position. When the retainer member **806** is rotated to the third position, the plug portion **710** can be inserted into and removed from the socket portion **805**.

[0185] FIGS. **42**A and **42**C are perspective views of the panel unit **801** in a state in which the plug portion **710** of the panel cable **703** has been inserted in the socket portion **805** of the panel unit **801**. FIGS. **42**B and **42**D correspond to FIGS. **42**A and **42**C, and are a cross-sectional view taken along a plane vertically extending through the screw member **816**, which passes the socket portion **805** and the screw member **816**, and a view of the retainer member **806** and the like as viewed from the inside of the retainer member **806**, respectively.

[0186] In FIGS. **42**A and **42**B, similar to FIG. **41**E, the retainer member **806** is in the third position. In this position, the cable engagement surface **811** of the retainer member **806** and the retainer member contact surface **728** of the grasping portion **722** are not brought into contact with each other, and hence it is possible to freely insert and remove the plug portion **710** of the panel cable **703**.

[0187] In FIGS. **42**C and **42**D, similar to FIG. **41**A, the retainer member **806** is in the first position. In this position, the cable engagement surface **811** and the retainer member contact surface **728** are brought into contact with each other, and hence the plug portion **710** is prevented from being pulled off from the socket portion **805**.

[0188] FIGS. **43**A and **43**C are perspective views of the panel unit **801** in a state in which the plug portion **710** of the panel cable **723** is inserted in the socket portion **805** of the panel unit **801**. FIGS. **43**B and **43**D correspond to FIGS. **43**A and **43**C, and are a cross-sectional view taken along a plane vertically extending through the screw member **816**, which passes the socket portion **805** and the screw member **816**, and a view of the retainer member **806** and the like, as viewed from the inside of the retainer member **806**, respectively.

[0189] As described above, the panel cable **723** is different in the length of the grasping portion from the panel cable **703**. Therefore, after the plug portion **710** is inserted, the retainer member **806** is set to the second position.

[0190] First, in FIGS. **43**A and **43**B, similar to FIG. **41**E, the retainer member **806** is in the third position. In this position, the cable engagement surface **811** of the retainer member **806** and the retainer member contact surface **729** of the grasping portion **723** are not brought into contact with each other, and hence it is possible to freely insert and remove the plug portion **710** of the panel cable **723**.

[0191] In FIGS. **43**C and **43**D, similar to FIG. **41**C, the retainer member **806** is in the second position. In this position, the cable engagement surface **811** and the retainer member contact surface **729** are in contact with each other, and hence the plug portion **710** is prevented from being pulled off from the socket portion **805**.

[0192] Thus, when the plug portion **710** is inserted into the socket portion **805**, the cable engagement surface **811** is engaged with the grasping portion (**722** or **725**) in the first position or the second position, whereby the function of retaining the plug portion **710** by the retainer member **806** is achieved.

[0193] According to the present embodiment, it is possible to obtain the
[0194] same advantageous effects as provided by the first embodiment with respect to improvement of the operability while reducing the size of the grip **111**.
[0195] Further, since the retainer member **806** can take the first position and the second position, it is possible to prevent the plug portions of the panel cables (**703** and **723**) different in the length of the grasping portions (**722** and **725**) from being pulled off from the socket portion **805**.
[0196] Note that although each cable described in the second and third embodiments is a panel cable for connecting between the image capturing apparatus body **104** and the panel unit, this is not limitative, but it can be a cable for connecting between the image capturing apparatus body **104** and a device other than the display device.
[0197] Note that in the second and third embodiments, the example of application of the structure for retaining the panel cable to the panel unit has been described. However, this is not limitative, and the same cable retaining structure can be applied to the image capturing apparatus body **104**.
[0198] Note that the present invention can also be applied to an image capturing apparatus formed by integrating the image capturing apparatus body **104** and the lens.
[0199] Note that in the above-described embodiments, an expression including “substantially” is not intended to exclude “completely”. For example, expressions of “substantially parallel”, “substantially perpendicular”, “substantially orthogonal”, “substantially the same”, “substantially rectangular shape”, and “substantially rectangular parallelepiped shape” are intended to include “completely parallel”, “completely perpendicular”, “completely orthogonal”, “completely the same state” “completely rectangular shape”, and “completely rectangular parallelepiped shape”, respectively.
Other Embodiments
[0200] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD) TM), a flash memory device, a memory card, and the like.
[0201] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0202] This application claims the benefit of Japanese Patent Application No. 2024-026566 filed Feb. 26, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image capturing apparatus comprising: an image capturing apparatus body; and a grip rotatably attached to a side surface of the image capturing apparatus body, wherein the grip includes: a grasping portion for being grasped by a user; a first, a second, and a third operation elements for being operated by the user; a first exterior that covers the first operation element at least from the image capturing apparatus body; and a second exterior that covers the third operation element at least from the image capturing apparatus body, wherein out of rotational positions of the grip, when the grip is in a first rotational position in which the first operation element and the second operation element are positioned upward of the grasping portion, the third operation element is positioned between the first operation element and the second operation element at a location upward of a line segment connecting between the first operation element and the second operation element, and at the same time, the second exterior is more remote from the image capturing apparatus body than the first exterior.
2. The image capturing apparatus according to claim 1, wherein in the image capturing apparatus body, an operation member for operating the image capturing apparatus body is disposed at a location opposed to the second exterior when the grip is in the first rotational position.
3. The image capturing apparatus according to claim 1, wherein the first, second, third operation elements are any of a push-type key, a direction input key, a slide key, a dial key, and a toggle key.
4. The image capturing apparatus according to claim 1, wherein the first and third operation elements are each a push-type key, wherein the second operation element is a direction input key, and wherein a distance between respective centers of the first operation element and the third operation element is shorter than both of a distance between respective centers of the first operation element and the second operation element and a distance between respective centers of the second operation element and the third operation element.
5. The image capturing apparatus according to claim 1, wherein the line segment is substantially parallel to a rotational axis of the grip for the image capturing apparatus body.
6. The image capturing apparatus according to claim 2, wherein the operation member is a member that prevents the cable from being pulled off from the image capturing apparatus body, when a cable for connecting between the image capturing apparatus body and a display device is inserted into the image capturing apparatus body.
7. The image capturing apparatus according to claim 1, wherein the grip is in a second rotational position different from the first rotational position, an air exhaust port provided in the image capturing apparatus body and the first exterior do not overlap each other as viewed from an axial direction of a rotational axis of the grip for the image capturing apparatus body.
8. The image capturing apparatus according to claim 1, wherein the side surface of the image capturing apparatus body is provided with an air exhaust port for exhausting air warmed inside the image capturing apparatus body, and wherein the air exhaust port is located in the side surface at an end opposite to an object side in a direction of an optical axis and at the same time upward of the optical axis, and opens in the side surface in a substantially perpendicular direction.
9. The image capturing apparatus according to claim 8, wherein the air exhaust port is disposed on a side opposite to an object in the optical axis direction and at the same time upward, with respect to an attaching portion of the image capturing apparatus body where the grip is attached.
10. The image capturing apparatus according to claim 9, wherein compared with the air exhaust port, the attaching portion more protrudes outward of the image capturing apparatus body in a direction perpendicular to the side surface.
11. The image capturing apparatus according to claim 8, wherein an air inlet port is disposed more toward the object side in the optical axis direction than the air exhaust port.
12. The image capturing apparatus according to claim 1, wherein the image capturing apparatus body includes a first socket portion into which a plug of a cable for connecting another device can be removably inserted, and a first retainer member, wherein the cable has a protruding portion,

wherein the first retainer member has a first engaging portion, and is at the same time rotatable about a first axis substantially parallel to a direction of insertion and removal of the plug into and from the first socket portion, and wherein when the plug is inserted into the first socket portion, the first engaging portion is engaged with the protruding portion, whereby a retaining function of the first retainer member for retaining the plug is achieved.

13. The image capturing apparatus according to claim 12, wherein an outer shape of the image capturing apparatus body has a substantially rectangular shape as viewed from an object side, and wherein the first socket portion and the first retainer member are disposed in one corner of the rectangular shape.

14. The image capturing apparatus according to claim 12, wherein the direction of insertion and removal of the plug is substantially parallel to the optical axis.

15. The image capturing apparatus according to claim 12, wherein a display device is communicably connected to the image capturing apparatus body by the cable.

16. The image capturing apparatus according to claim 12, including a display device communicably connected to the image capturing apparatus body by the cable, wherein the display device includes a second socket portion into which a plug of the cable can be removably inserted, and a second retainer member, wherein the second retainer member includes a second engagement portion, and at the same time, is rotatable about a second axis substantially parallel to a direction of insertion and removal of the plug to and from the second socket portion, and wherein when the plug is inserted into the second socket portion, the second engaging portion is engaged with the protruding portion, whereby a retaining function of the second retainer member for retaining the plug is achieved.

17. The image capturing apparatus according to claim 16, wherein the second socket is provided in a rear surface of the display device, and wherein the direction of insertion and removal of the plug to and from the second socket portion is substantially parallel to the rear surface of the display device.

18. The image capturing apparatus according to claim 12, wherein the cable includes two grasping portions, and a flexible member to which the second retainer is connected, and wherein the protruding portion is disposed at each of opposite ends of the flexible member.

19. The image capturing apparatus according to claim 12, wherein the protruding portion is provided such that the protruding portion protrudes substantially perpendicular to the direction of insertion and removal of the plug.

20. The image capturing apparatus according to claim 12, wherein the image capturing apparatus body is provided with a recess shaped portion for avoiding the protruding portion when the plug is inserted into the first socket portion, and wherein the recess shaped portion has a shorter length in the direction of insertion and removal of the plug than a length of engagement between the plug and the first socket portion.

21. The image capturing apparatus according to claim 20, wherein the cable includes a grasping portion, and wherein the protruding portion is disposed on each of opposite side surfaces of the grasping portion in a direction perpendicular to the direction of insertion and removal of the plug.

22. The image capturing apparatus according to claim 12, wherein the cable includes a grasping portion, and wherein the image capturing apparatus body is provided with a protruding shape portion brought into contact with the grasping portion when the plug is inserted into the first socket portion.

23. The image capturing apparatus according to claim 12, wherein the image capturing apparatus body is provided with a cover portion engageable with the first socket portion, and wherein the first retainer member is fixed to the image capturing apparatus body in a state in which the cover portion is engaged with the first socket portion, whereby movement of the cover portion is restricted.

24. The image capturing apparatus according to claim 16, wherein the display device is provided with a recess shaped portion for avoiding the protruding portion when the plug is inserted into the second socket portion, and wherein the recess shaped portion has a longer length in the direction of insertion and removal of the plug than a length of engagement between the plug and the second socket portion.

25. The image capturing apparatus according to claim 12, wherein the cable includes a grasping portion, and wherein the display device is provided with a protruding shape portion brought into contact with the grasping portion when the plug is inserted into the second socket portion.

26. The image capturing apparatus according to claim 1, wherein the image capturing apparatus body includes a socket portion into which a plug of the cable for connecting another device can be removably inserted, a retainer member, and a fixing member wherein the retainer member has an engaging portion, wherein the retainer member is slidable in the direction of insertion and removal of the plug between the first position and a second position more remote from the socket portion than the first position, and at the same time, is rotatable to a third position in which the engaging portion retreats from between the first position and the second position, wherein the retainer member is rotated to the third position, whereby the plug can be inserted into and removed from the socket portion, wherein when the plug is inserted into the socket portion, the engaging portion is engaged with a cable grasping portion which the cable includes, at the first position or at the second position, whereby a retaining function of retaining the plug by the retainer member is achieved, and wherein the retainer member is fixed to the image capturing apparatus body by the fixing member, whereby sliding movement of the retainer member is restricted.

27. The image capturing apparatus according to claim 26, wherein the fixing member restricts the sliding movement of the retainer member, by urging part of the retainer member toward the image capturing apparatus body.

28. The image capturing apparatus according to claim 27, wherein the fixing member includes a screw member, and the retainer member is fixed to the image capturing apparatus body by screwing the screw member.

29. The image capturing apparatus according to claim 26, wherein the retainer member rotates about a rotational axis substantially parallel to a sliding direction, whereby the retainer member is rotatable to the third position.

30. The image capturing apparatus according to claim 26, wherein the retainer member rotates about a rotational axis substantially orthogonal to a sliding direction, whereby the retainer member is rotatable to the third position.

31. The image capturing apparatus according to claim 1, further comprising a display device communicably connected to the image capturing apparatus body by a cable, and wherein the display device includes a socket portion into which a plug of the cable can be removably inserted, a retainer member, and a fixing member, wherein the retainer member has an engaging portion, wherein the retainer member is slidable in the direction of insertion and removal of the plug between the first position and a second position more remote from the socket portion than the first position, and at the same time, is a rotatable to a third position in which the engaging portion retreats from between the first position and the second position, wherein the retainer member is rotated to the third position, whereby the plug can be inserted into and removed from the socket portion, wherein when the plug is inserted into the socket portion, the engaging portion is engaged with a cable grasping portion which the cable includes, at the first position or at the second position, whereby a retaining function of retaining the plug by the retainer member is achieved, and wherein the retainer member is fixed to the image capturing apparatus body by the fixing member, whereby sliding movement of the retainer member is restricted.

32. The image capturing apparatus according to claim 31, wherein the fixing member restricts the sliding movement of the retainer member, by urging part of the retainer member toward the image capturing apparatus body.

33. The image capturing apparatus according to claim 32, wherein the fixing member includes a screw member, and the retainer member is fixed to a body of the display device by screwing the screw member.

34. The image capturing apparatus according to claim 31, wherein the retainer member rotates about a rotational axis substantially parallel to a sliding direction, whereby the retainer member is rotatable to the third position.

35. The image capturing apparatus according to claim 31, wherein the retainer member rotates about a rotational axis substantially orthogonal to a sliding direction, whereby the retainer member is rotatable to the third position.